

Ê PHASOR TEMPLATES (Air: ε<sub>r</sub>=μ<sub>r</sub>=1)

Wavenumber per medium *nonmag.*

$k_i = \frac{\omega\sqrt{\epsilon_{r,i}}}{c}$  (rad/m)  $\Rightarrow k_2 = k_1\sqrt{\epsilon_{r,2}/\epsilon_{r,1}}$

Direction table  $u \in \{x, y, z\} = \text{axis perp. to boundary}$

| Inc dir | Inc           | Refl          | Trans         |
|---------|---------------|---------------|---------------|
| +û      | $e^{-jk_1 u}$ | $e^{+jk_1 u}$ | $e^{-jk_2 u}$ |
| -û      | $e^{+jk_1 u}$ | $e^{-jk_1 u}$ | $e^{+jk_2 u}$ |

+û direction ⇔ Med 2 at u ≥ 0. Reflected sign flips, transmitted matches incident.

General template (incident has pol. ê, amplitude a<sub>0</sub>):

$\hat{\mathbf{E}}^i = a_0 \hat{e} e^{\mp j k_1 u}$  (incident wave, Med 1)

$\hat{\mathbf{E}}^r = \Gamma a_0 \hat{e} e^{\pm j k_1 u}$  (reflected wave, Med 1)

$\hat{\mathbf{E}}^t = \tau a_0 \hat{e} e^{\mp j k_2 u}$  (transmitted wave, Med 2)

(lossy med 2: replace  $e^{\mp j k_2 u}$  with  $e^{\mp \alpha_2 u} e^{\mp j \beta_2 u}$ )

Total in medium 1:  $\mathbf{E}_1 = \hat{\mathbf{E}}^i + \hat{\mathbf{E}}^r$ .

ê by polarization *plug into templates above*

| Pol.        | ê                                                    | Note               |
|-------------|------------------------------------------------------|--------------------|
| Lin. x̂     | x̂                                                   | E <sub>y</sub> = 0 |
| Lin. ŷ      | ŷ                                                    | E <sub>x</sub> = 0 |
| Lin. at 45° | (x̂ + ŷ)/√2                                          | in phase           |
| RHC         | x̂ - jŷ                                              | δ = -π/2           |
| LHC         | x̂ + jŷ                                              | δ = +π/2           |
| General     | a <sub>x</sub> x̂ + a <sub>y</sub> e <sup>jδ</sup> ŷ | elliptical         |

Γ, τ act on ê unchanged. Only swap the polarization, not the formulas.

POWER REFLECTION & TRANSMISSION

% reflected *always*

$\%P_r = 100 |\Gamma|^2$

% transmitted *η-ratio matters!*

$\%P_t = 100 |\tau|^2 \frac{\eta_1}{\eta_2}$

Check: %P<sub>r</sub> + %P<sub>t</sub> = 100.

If medium 2 is denser (η<sub>2</sub> < η<sub>1</sub>), |τ| > 1 but %P<sub>t</sub> < 100% because of the η<sub>1</sub>/η<sub>2</sub> factor.

LOSSY MEDIA — 5 CASES

Loss tangent  $\epsilon' = \epsilon_r \epsilon_0, \epsilon_0 \approx 1/(36\pi \times 10^9)$

$\frac{\epsilon''}{\epsilon'} = \frac{\sigma}{\omega \epsilon_r \epsilon_0}$

| Type         | σ/ωε                               | ηc, α, β                                                                                                                                                                             |
|--------------|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Lossless     | 0                                  | η = η <sub>0</sub> /√ε <sub>r</sub> exact; α = 0, β = ω√ε <sub>r</sub> /c                                                                                                            |
| Low-loss     | < 10 <sup>-2</sup>                 | ηc ≈ $\frac{\eta_0}{\sqrt{\epsilon_r}} \left(1 + j \frac{\sigma}{2\omega\epsilon'}\right)$ ; α ≈ $\frac{\sigma\eta_0}{2\sqrt{\epsilon_r}}$ , β ≈ $\frac{\omega\sqrt{\epsilon_r}}{c}$ |
| Quasi-cond.  | 10 <sup>-2</sup> – 10 <sup>2</sup> | exact: (η/√X)e <sup>jθ</sup> η, see below                                                                                                                                            |
| Good cond.   | > 10 <sup>2</sup>                  | ηc ≈ (1+j)√πfμ/σ; α ≈ β ≈ √πfμσ                                                                                                                                                      |
| Magnetic any |                                    | keep full √μ <sub>r</sub> /ε <sub>r</sub> η <sub>0</sub>                                                                                                                             |

For Γ/τ: drop j correction; use η<sub>0</sub>/√ε<sub>r</sub>. Magnetic: keep μ<sub>r</sub>.

Exact (quasi-cond)  $X = \sqrt{1 + (\epsilon''/\epsilon')^2}$

$\alpha = \omega\sqrt{\frac{\mu\epsilon'}{2}}\sqrt{X-1}$

$\beta = \omega\sqrt{\frac{\mu\epsilon'}{2}}\sqrt{X+1}$

$\theta_\eta = \frac{1}{2} \arctan \frac{\sigma}{\omega\epsilon'}$

SNELL'S LAW — OBLIQUE INCIDENCE

Single boundary *angles from normal*

$\sin \theta_2 = \sin \theta_1 \sqrt{\frac{\epsilon_{r1}}{\epsilon_{r2}}} = \sin \theta_1 \frac{n_1}{n_2}$

Multilayer chain *stack of slabs 1 → 2 → 3 → 4*

$\sin \theta_{i+1} = \sin \theta_i \sqrt{\frac{\epsilon_{r,i}}{\epsilon_{r,i+1}}}$

Air-slabs-air shortcut *outer media identical*

$\theta_{\text{exit}} = \theta_{\text{incident}}$

Snell only sets angles. Reflection/transmission magnitudes need Fresnel coefficients (parallel vs perpendicular).

LATERAL DISPLACEMENT

Beam offset through N slabs thickness t<sub>i</sub>, refracted angle inside slab i

Slab-indexed θ<sub>i</sub> = angle inside slab i

$d = \sum_{i=1}^N t_i \tan \theta_i$

Angle-indexed (HW3 P3 style) θ<sub>1</sub> = incident, θ<sub>i+1</sub> = inside slab i

$d = \sum_{i=1}^N t_i \tan \theta_{i+1}$

Same physics, different numbering. In HW3 P3: slab 1 contributes t<sub>1</sub> tan θ<sub>2</sub>, slab 2 contributes t<sub>2</sub> tan θ<sub>3</sub>. The angle in the formula is always the one the ray makes inside that slab.

Air-slab-air: exit angle returns to θ<sub>1</sub> but is laterally shifted by d.

TABLE 8-2 — Γ, τ, R, T SUMMARY

Med. 1 (z < 0) incident side; Med. 2 (z ≥ 0) transmitted.

Shorthand: c<sub>i</sub> = cos θ<sub>i</sub>, c<sub>t</sub> = cos θ<sub>t</sub>.

|               | Normal (θ <sub>i</sub> =0)                | ⊥ pol                                                         | ∥ pol                                                         |
|---------------|-------------------------------------------|---------------------------------------------------------------|---------------------------------------------------------------|
| Γ refl. wave  | $\frac{\eta_2 - \eta_1}{\eta_2 + \eta_1}$ | $\frac{\eta_2 c_i - \eta_1 c_t}{\eta_2 c_i + \eta_1 c_t}$     | $\frac{\eta_2 c_t - \eta_1 c_i}{\eta_2 c_t + \eta_1 c_i}$     |
| τ trans. wave | $\frac{2\eta_2}{\eta_2 + \eta_1}$         | $\frac{2\eta_2 c_i}{\eta_2 c_i + \eta_1 c_t}$                 | $\frac{2\eta_2 c_t}{\eta_2 c_t + \eta_1 c_i}$                 |
| τ rel.        | τ = 1 + Γ                                 | τ <sub>⊥</sub> = 1 + Γ <sub>⊥</sub>                           | τ <sub>∥</sub> = (1 + Γ <sub>∥</sub> ) $\frac{c_i}{c_t}$      |
| R refl. pwr   | Γ  <sup>2</sup>                           | Γ <sub>⊥</sub>   <sup>2</sup>                                 | Γ <sub>∥</sub>   <sup>2</sup>                                 |
| T trans. pwr  | τ  <sup>2</sup> $\frac{\eta_1}{\eta_2}$   | τ <sub>⊥</sub>   <sup>2</sup> $\frac{\eta_1 c_t}{\eta_2 c_i}$ | τ <sub>∥</sub>   <sup>2</sup> $\frac{\eta_1 c_t}{\eta_2 c_i}$ |
| R + T total   | T = 1 - R                                 | T <sub>⊥</sub> = 1 - R <sub>⊥</sub>                           | T <sub>∥</sub> = 1 - R <sub>∥</sub>                           |

Notes: sin θ<sub>t</sub> = √μ<sub>1</sub>ε<sub>1</sub>/μ<sub>2</sub>ε<sub>2</sub> sin θ<sub>i</sub>; nonmag. ⇒ η<sub>2</sub>/η<sub>1</sub> = n<sub>1</sub>/n<sub>2</sub>.

Intrinsic impedance η<sub>i</sub> *plug into table above*

$\eta_i = \sqrt{\frac{\mu_r}{\epsilon_r}} \eta_0 \xrightarrow{\mu_r=1} \frac{\eta_0}{\sqrt{\epsilon_{r,i}}} \text{ (}\Omega\text{)} \quad \eta_0 = 120\pi \approx 377 \Omega$

Default μ<sub>r</sub> = 1 (air, dielectric). Use μ<sub>r</sub> ≠ 1 only if problem says “ferromagnetic”/iron/ferrite/etc.

Low-loss correction η<sub>c</sub> σ/(ωε) ≪ 1

$\eta_c \approx \sqrt{\frac{\mu}{\epsilon'}} \left(1 + j \frac{\sigma}{2\omega\epsilon'}\right) \xrightarrow{\text{drop } j} \frac{\eta_0}{\sqrt{\epsilon_r}}$

For Γ/τ just use η<sub>0</sub>/√ε<sub>r</sub>. The j term is for |η<sub>c</sub>|, θ<sub>η</sub>. See LOSSY MEDIA — 5 CASES for all regimes.

RECTANGULAR WAVEGUIDE — MODES

TE mode E<sub>z</sub> = 0, H<sub>z</sub> ≠ 0

TM mode H<sub>z</sub> = 0, E<sub>z</sub> ≠ 0

Dimensions a × b with a > b along x̂, ŷ; wave in ẑ.

E<sub>x</sub> form (TE) *read m, n from arguments*

$E_x \propto \cos\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right) \sin(\omega t - \beta z)$

coef. on x = mπ/a; coef. on y = nπ/b.

Identify TE<sub>m,n</sub> vs TM<sub>m,n</sub> *cos/sin pattern of E<sub>x</sub> is identical for both!*

1. Read the problem statement (e.g. “a TE wave” → TE).

2. If m = 0 or n = 0 in the mode label ⇒ must be TE (TM forbids 0).

3. Source field H<sub>z</sub>(x, y) given ⇒ TE. Source field E<sub>z</sub>(x, y) given ⇒ TM.

TE allows at least one of m, n ≥ 1 (other can be 0). TM needs both m, n ≥ 1. That's why TE<sub>10</sub> exists but TM<sub>10</sub> doesn't.

TABLE 8-3 — FULL FIELD COMPONENTS

Shorthand: C<sub>X</sub> = cos  $\frac{m\pi x}{a}$ , S<sub>X</sub> = sin  $\frac{m\pi x}{a}$ , C<sub>Y</sub> = cos  $\frac{n\pi y}{b}$ , S<sub>Y</sub> = sin  $\frac{n\pi y}{b}$ . All fields carry an e<sup>-jβz</sup> factor (omitted below). k<sub>c</sub><sup>2</sup> = (mπ/a)<sup>2</sup> + (nπ/b)<sup>2</sup>.

TE mode generator:  $\hat{H}_z$

$\hat{H}_z = H_0 C_X C_Y$

$\hat{E}_x = \frac{j\omega\mu}{k_c^2} \left(\frac{n\pi}{b}\right) H_0 C_X S_Y$

$\hat{E}_y = -\frac{j\omega\mu}{k_c^2} \left(\frac{m\pi}{a}\right) H_0 S_X C_Y$

$\hat{E}_z = 0, \quad \hat{H}_x = -\hat{E}_y/Z_{TE}, \quad \hat{H}_y = \hat{E}_x/Z_{TE}$

TM mode generator:  $\hat{E}_z$

$\hat{E}_z = E_0 S_X S_Y$

$\hat{E}_x = -\frac{j\beta}{k_c^2} \left(\frac{m\pi}{a}\right) E_0 C_X S_Y$

$\hat{E}_y = -\frac{j\beta}{k_c^2} \left(\frac{n\pi}{b}\right) E_0 S_X C_Y$

$\hat{H}_z = 0, \quad \hat{H}_x = -\hat{E}_y/Z_{TM}, \quad \hat{H}_y = \hat{E}_x/Z_{TM}$

TEM (plane wave) for reference:  $\hat{E}_x = E_0 e^{-j\beta z}, \hat{H}_y = \hat{E}_x/\eta, f_c \text{ N/A}, k = \omega\sqrt{\mu\epsilon}$ .

CUTOFF FREQUENCY

f<sub>m,n</sub> *m, n integers ≥ 0 (not both zero)*

$f_{mn} = \frac{u_{p0}}{2} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2} \text{ (Hz)}$

Common cutoffs (a > b)

| Quantity                            | Formula                                         |
|-------------------------------------|-------------------------------------------------|
| u <sub>p0</sub>                     | c/√ε <sub>r</sub> (hollow: u <sub>p0</sub> = c) |
| TE <sub>10</sub>                    | u <sub>p0</sub> /(2a) (dominant)                |
| TE <sub>01</sub>                    | u <sub>p0</sub> /(2b)                           |
| TE <sub>20</sub>                    | u <sub>p0</sub> /a = 2f <sub>10</sub>           |
| TE <sub>11</sub> , TM <sub>11</sub> | $\frac{u_{p0}}{2} \sqrt{1/a^2 + 1/b^2}$         |

Propagation requires f > f<sub>m,n</sub>.

ε<sub>r</sub>, FROM DISPERSION

Given ω, β, mode (m, n)

$\epsilon_r = \frac{c^2}{\omega^2} \left[ \beta^2 + \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2 \right]$

Sanity: β in rad/m, a, b in m, ω in rad/s.

WAVE IMPEDANCE & H<sub>y</sub>

$Z_{TE} = \frac{\eta}{\sqrt{1 - (f_c/f)^2}} \text{ (}\Omega\text{)} \quad Z_{TM} = \eta \sqrt{1 - (f_c/f)^2}$

$\eta = \frac{\eta_0}{\sqrt{\epsilon_r}} \quad H_y = \frac{E_x}{Z_{TE}} \text{ (TE mode)}$

Z<sub>TE</sub> > η always (denominator < 1); Z<sub>TM</sub> < η.

GROUP VELOCITY & TRAVEL TIME

$u_g = u_{p0} \sqrt{1 - (f_{mn}/f)^2} \text{ (m/s)}$

$u_p = \frac{u_{p0}}{\sqrt{1 - (f_{mn}/f)^2}} \text{ (m/s)} \Rightarrow u_p u_g = u_{p0}^2 \text{ (m}^2\text{/s}^2\text{)}$

t = L/u<sub>g</sub> (s) travel time through guide of length L

Higher modes have lower u<sub>g</sub> ⇒ they arrive later. Group delay spread = sum over excited modes.

BOUNDARY

η<sub>1</sub>, η<sub>2</sub> — imp. (Ω)

Γ — refl. coeff (unitless)

τ = 1 + Γ — trans. coeff

k<sub>i</sub> = ω√ε<sub>r,i</sub>/c (rad/m)

α<sub>2</sub>, β<sub>2</sub> — if med 2 lossy

z = 0 — boundary plane

η<sub>0</sub> = 120π ≈ 377 Ω

SNELL / OBLIQUE

θ<sub>i</sub>, θ<sub>t</sub> — angles from normal

n<sub>i</sub> = √ε<sub>r,i</sub> — index of refr.

t<sub>i</sub> — slab thickness (m)

d — lateral disp. (m)

plane of inc. = plane of k̂ + normal

⊥: **E** ⊥ this plane

∥: **E** ∥ this plane

WAVEGUIDE

a, b — wide, narrow dim. (m); a > b

m, n — mode indices (int.)

f<sub>mn</sub> — cutoff (Hz)

β — prop. const (rad/m)

Z<sub>TE</sub>/Z<sub>TM</sub> — wave imp. (Ω)

E<sub>x</sub>, H<sub>y</sub> — field comps (V/m, A/m)

η = η<sub>0</sub>/√ε<sub>r</sub>

VELOCITIES

u<sub>p0</sub> = c/√ε<sub>r</sub> — bulk phase vel.

u<sub>p</sub> = u<sub>p0</sub>/√1 - (f<sub>c</sub>/f)<sup>2</sup>

u<sub>g</sub> = u<sub>p0</sub>√1 - (f<sub>c</sub>/f)<sup>2</sup>

u<sub>p</sub> u<sub>g</sub> = u<sub>p0</sub><sup>2</sup>

t = L/u<sub>g</sub> — travel time

c = 3 × 10<sup>8</sup> m/s

POWER & UNITS

%P<sub>r</sub> = 100|Γ|<sup>2</sup>

%P<sub>t</sub> = 100|τ|<sup>2</sup>η<sub>1</sub>/η<sub>2</sub>

%P<sub>r</sub> + %P<sub>t</sub> = 100

σ (S/m), ε (F/m), μ (H/m)

ε<sub>0</sub> = 8.85 × 10<sup>-12</sup> F/m

μ<sub>0</sub> = 4π × 10<sup>-7</sup> H/m

1/(36π) × 10<sup>-9</sup> ≈ ε<sub>0</sub>